

MINE SENTINEL AI

INTELLIGENT IOT-BASED COAL MINE SAFETY MONITORING AND PREDICTIVE ALERT SYSTEM USING ESP32

BATCH NUMBER: WI 12

PRIMARY DOMAIN: EDGE IOT + INDUSTRIAL IOT (IIoT) + MACHINE LEARNING + PREDICTIVE ANALYTICS

BASE RESEARCH PAPER: 'IoT-Based Coal Mine Safety Monitoring and Alerting System'

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EXECUTIVE SUMMARY & ABSTRACT

Mine Sentinel AI is an intelligent, cyber-physical Edge IoT and Industrial IoT (IIoT) safety monitoring system engineered to safeguard underground coal mine workers against volatile environmental hazards. Underground coal mining represents one of the most perilous industrial sectors globally, where catastrophic perils such as explosive methane (CH₄) accumulation, lethal carbon monoxide (CO) asphyxiation, spontaneous combustion fires, and thermal heat-stress claim dozens of miner lives every year. Conventional monitoring practices rely on static handheld detectors or rudimentary threshold alarms that frequently fail to identify multi-parameter compounding hazards until conditions reach irreversible, life-threatening critical points.

To decisively solve these challenges, Mine Sentinel AI integrates a dual-core 32-bit **ESP32 NodeMCU** microcontroller with an external **ADS1115 16-bit Delta-Sigma ADC**, high-sensitivity **MQ-2 Combustible Gas**, **MQ-7 Carbon Monoxide**, **DHT11 Temperature & Humidity**, and **Infrared Optical Flame** sensors. The edge node features a local **16x2 Character I2C LCD** display, delivering zero-latency situational awareness directly to subterranean personnel at the coal face, even in the event of total underground network severance. Live telemetry is transmitted via lightweight **MQTT** over Wi-Fi to a high-throughput **Python FastAPI** cloud backend, which evaluates incoming sensor vectors through an optimized **Scikit-Learn Random Forest Classifier**.

The system classifies mine safety into three actionable states: **Safe**, **Warning**, and **Critical**. When hazardous conditions or predictive fire trends are detected, Mine Sentinel AI initiates autonomous physical mitigation by energizing an optocoupled relay to drive a **12V industrial exhaust ventilation fan**, actuating a high-decibel active buzzer, and illuminating tri-color LEDs. Simultaneously, live spatial telemetry is rendered onto a responsive Web Supervisory Dashboard powered by **React 18 Modular Islands** (CopilotChat, FleetNodesGrid, TelemetryCards, AlertCenter, ShiftAuditModal) with authenticated role-based security ('Get Started' workflow), accompanied by an automated **ReportLab statutory shift compliance engine** that generates official DGMS Form IV / MSHA preshift safety examination PDF audits. With an empirical **98.00% classification accuracy** and sub-second end-to-end response latency, the system establishes a low-cost, scalable Industry 4.0 standard for hazardous industrial surveillance.

Core Innovation	Hybrid Edge-Cloud Machine Learning with Autonomous Closed-Loop Relay Ventilation Actuation
Edge Controller	ESP32 NodeMCU (30-Pin) + ADS1115 16-bit ADC + Decoupled Anti-Brownout Power Network
Local Display	16x2 Character I2C LCD (PCF8574 @ 0x27) providing subterranean situational awareness
Presentation & UI	React 18 Modular Islands (CopilotChat, FleetNodesGrid, AlertCenter) + Real-Time Chart.js
ML Model & Metric	Random Forest (100 Trees) 98.00% Test Accuracy Weighted F1: 0.9793
Statutory Audit	Automated DGMS CMR 2017 Reg. 153/154 & MSHA 30 CFR Section 75.360 PDF Compliance Engine

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CHAPTER 1: INTRODUCTION, PROBLEM STATEMENT & OBJECTIVES

1.1 Background and Context

Underground coal extraction remains one of the world's most severe operational environments. Miners perform labor deep below the surface in narrow galleries where catastrophic risks are ever-present: methane gas pockets released from freshly cut coal faces, lethal carbon monoxide emitted from sub-surface oxidation, rapid oxygen depletion, extreme thermal stress, and the immediate threat of open flames or explosive dust clouds. Because human sensory organs cannot perceive colorless, odorless gases such as methane or carbon monoxide, miners are entirely dependent on electronic safety instruments to avert disaster.

1.2 The Base Research Paper

This project builds upon and substantively advances the research presented in the IEEE conference publication: **'IoT-Based Coal Mine Safety Monitoring and Alerting System'** (authored by Harivardhagini Subhadra and Sreelatha Reddy Vakiti, published in *2024 International Conference on Social and Sustainable Innovations in Technology and Engineering [SASI-ITE]*, IEEE Xplore, DOI: 10.1109/SASI-ITE58663.2024.00051). While the base paper validated the feasibility of rudimentary sensor logging over wireless networks, it highlighted critical vulnerabilities in existing literature: reliance on primitive thresholding, high cloud latency, total lack of local visual feedback at the mine face, and absence of closed-loop automated ventilation.

1.3 Formal Problem Statement

Coal mining is severely hindered by the inability of traditional monitoring setups to provide continuous, high-precision, predictive hazard classification. Conventional safety monitoring systems rely on manual handheld inspections or basic threshold-based alarms that evaluate sensor metrics as isolated scalar values. Under dynamic subterranean conditions, disasters rarely emerge from single spikes; rather, compound interactions - such as a moderate rise in carbon monoxide coupled with elevated temperature and reduced humidity - indicate active smoldering long before static alarms trip. Furthermore, standard microcontrollers suffer from severe ADC non-linearities and RF noise, legacy cloud backends introduce unviable multi-second transmission latencies, subterranean workers lack immediate on-site visual displays, and manual shift compliance logs are prone to retrospective fabrication. Hence, there is an urgent necessity for an integrated cyber-physical edge AI platform that combines precision digitization, on-site visual monitoring, predictive ML risk classification, autonomous physical exhaust mitigation, and certified statutory compliance reporting.

1.4 Objectives of the Project

- **Continuous Multi-Parameter Environmental Acquisition:** Deploy calibrated MQ-2 (Combustible Gas), MQ-7 (Carbon Monoxide), DHT11 (Temperature & Humidity), and Optical Flame sensors to continuously monitor underground atmospheric variables.
- **High-Precision 16-Bit Signal Conditioning:** Eliminate microcontroller ADC distortion and RF noise by integrating an external ADS1115 16-bit Delta-Sigma ADC module with dual-rail power decoupling.
- **Immediate On-Site Subterranean Visual Feedback:** Render real-time gas telemetry, network connection status, and emergency warnings directly onto a ruggedized 16x2 Character I2C LCD screen at the coal face.
- **Asynchronous Cloud Ingestion & Edge-to-Cloud Telemetry:** Stream sensor payloads over Wi-Fi using the lightweight MQTT protocol into a high-throughput Python FastAPI backend microservice.
- **Multivariate Predictive Hazard Classification:** Train and deploy an optimized Scikit-Learn Random Forest Classifier (100 decision trees) to classify atmospheric safety into Safe, Warning, and Critical states with >95% accuracy.
- **Autonomous Closed-Loop Mitigation:** Automatically energize an optocoupled relay to drive a 12V industrial exhaust ventilation fan, sound a 5V active buzzer, and switch tri-color LEDs within milliseconds of hazard detection.
- **Automated Statutory Regulatory Compliance:** Integrate a vector-sharp ReportLab PDF engine to generate authenticated 2-page DGMS Form IV / MSHA 30 CFR Section 75.360 shift examination audit certificates directly from database telemetry.

CHAPTER 2: LITERATURE REVIEW, GAP ANALYSIS & NOVELTY

2.1 Critical Limitations of Existing Systems

- **Rigid Single-Variable Thresholding:** Conventional industrial nodes trigger alarms only when a single parameter crosses a hardcoded cutoff. They cannot detect non-linear compound hazards, such as simultaneous moderate elevations in CO and thermal index that indicate subterranean coal smoldering.
- **Microcontroller ADC Distortion & RF Noise:** Standard microcontrollers (e.g. ESP32 internal 12-bit SAR ADC) suffer from severe non-linearity near voltage rails and high-frequency noise induced by Wi-Fi transmissions, corrupting delicate analog gas sensor voltages.
- **Headless Operation (No Subterranean Display):** Most existing IoT prototypes stream data only to remote surface offices, leaving underground miners unaware of atmospheric toxicity. If network cables are damaged, miners face silent gas accumulation without warning.
- **High Latency & Cloud Log Bottlenecks:** Reliance on generic third-party platforms (e.g., ThingSpeak) introduces 15-second polling delays, lacks custom machine learning pipelines, and prevents dynamic risk interventions.
- **Passive Monitoring without Active Mitigation:** Conventional setups sound alarms but cannot autonomously intervene. They require manual dispatch of ventilation operators, permitting explosive gas pockets to spread before ventilation starts.
- **Manual Paper-Based Statutory Shift Auditing:** Mining regulations mandate preshift atmospheric checks every 8 hours. Paper logbooks are prone to clerical omission, transcription errors, and retrospective falsification.

2.2 Research Gaps Identified in Literature

- **Research Gap 1 - Multi-Sensor Fusion with Machine Learning:** Published literature rarely integrates supervised predictive classifiers into live telemetry ingestion. Scalar thresholds dominate, leaving compound micro-climate correlations unexploited.
- **Research Gap 2 - Disconnect Between Local Edge and Cloud Telemetry:** Systems either deploy purely offline buzzers without cloud records, or stream to cloud dashboards without any local visual interface. A resilient dual-tier topology is absent.
- **Research Gap 3 - Absence of Closed-Loop Autonomous Actuation:** Academic research focuses heavily on advisory alerts (SMS/Email) while ignoring low-latency physical mitigation like automated relay exhaust ventilation.
- **Research Gap 4 - Regulatory Disconnect:** Academic prototypes produce raw CSV dumps but fail to generate authenticated statutory compliance reports required by government safety directorates.

2.3 Proposed Innovations and Novelty in Mine Sentinel AI

Innovative Dimension	Conventional Systems / Base Paper	Mine Sentinel AI (Proposed Solution)
Predictive Hazard Classification	Static single-sensor cutoff thresholds; frequent false trips or delayed alarms.	Scikit-Learn Random Forest Classifier evaluating multivariate vectors in real time (98.0% accuracy).
Analog Signal Fidelity	Internal 12-bit ADC with severe non-linearity and Wi-Fi RF noise corruption.	External ADS1115 16-bit Delta-Sigma ADC with decoupled dual-rail capacitor power filter (0.125 mV resolution).
Local Situational Awareness	Headless nodes; no local screen; miners unaware of underground gas buildup.	On-site 16x2 Character I2C LCD rendering live PPM, thermal metrics, Wi-Fi status, and emergency alerts.
Hazard Mitigation	Passive advisory alarms only; relies on delayed human intervention.	Autonomous Closed-Loop Actuation: Optocoupled relay immediately powers 12V exhaust fan to purge fumes.
Backend Architecture	Generic third-party clouds (ThingSpeak) with 15s latency and no ML integration.	Custom asynchronous Python FastAPI backend + lightweight MQTT broker + SQLite database.
Regulatory Governance	Manual paper logbooks; no verification; susceptible to retrospective fabrication.	Automated ReportLab PDF compliance engine generating official DGMS Form IV / MSHA shift audits.

CHAPTER 3: SYSTEM ARCHITECTURE, DATA FLOW & TECH STACK

3.1 Comprehensive 8-Layer Architectural Stack

Mine Sentinel AI employs a structured, highly decoupled 8-layer cyber-physical architectural hierarchy. Each layer encapsulates specific physical, telemetric, analytical, or presentation functions, ensuring fault tolerance, sub-second execution latencies, and modular extensibility.

#	Layer Name	Core Components	Functional Responsibility
1	Sensing Layer	MQ-2, MQ-7, DHT11, Flame	Continuously senses combustible gas, CO, temp, humidity, and fire.
2	Edge Processing	ESP32 + ADS1115 16-Bit ADC	Digitizes analog signals at 16-bit, packages telemetry, handles fail-safe.
3	Communication	Wi-Fi 802.11 b/g/n, MQTT / REST	Transmits encrypted lightweight JSON packets to cloud broker.
4	Cloud / Backend	Python FastAPI & Uvicorn	Asynchronously ingests telemetry streams and handles REST API routes.
5	AI Prediction	Random Forest (Scikit-Learn)	Evaluates 5-dimensional vectors to predict risk (Safe/Warning/Critical).
6	Database Layer	SQLite + SQLAlchemy ORM	ACID-compliant storage of historical telemetry, alerts, and shift logs.
7	Presentation	React 18 Islands, HTML5/CSS3/JS, Chart.js & 16x2 LCD	Supervisory command center with React 18 modular islands, live Chart.js curves, AI copilot, and local LCD display.
8	Alert & Actuation	Relay Fan, Buzzer, Tri-Color LEDs	Closed-loop emergency exhaust actuation and multi-tier audiovisual alarms.



Fig 3.1: Detailed 8-Layer Cyber-Physical Architectural Diagram of Mine Sentinel AI

3.2 End-to-End Data Telemetry Pipeline

The telemetry pipeline operates continuously across milliseconds. Physical atmospheric phenomena are transduced by the sensor cluster, digitized via the 16-bit ADS1115 ADC, evaluated by the ESP32 edge processor, rendered locally on the 16x2 LCD, and broadcast via MQTT to the FastAPI cloud backend. The backend executes AI inference, persists the event to SQLite, updates the supervisory dashboard, and triggers closed-loop mitigation (exhaust fan & buzzer) whenever risks escalate.



Fig 3.2: Technical Workflow and End-to-End Telemetry Ingestion Pipeline

3.3 Comprehensive Technology Stack

Domain / Layer	Technology / Tool	Version / Specifications	Role in Architecture
Edge IoT	ESP32 NodeMCU	30-Pin, Tensilica Xtensa Dual-Core	Edge data acquisition, LCD control, Wi-Fi MQTT publisher.
ADC Subsystem	ADS1115 I2C ADC	16-bit Delta-Sigma, PGA +/-4.096V	Ultra-precise analog digitization for MQ-2 & MQ-7 sensors.
Local Display	16x2 Character LCD	JHD 162A + PCF8574 I2C (0x27)	On-site visual feedback for miners at the coal face.
Backend	FastAPI + Uvicorn	Python 3.11, Asynchronous REST	High-throughput telemetry ingestion & API orchestration.
Message Broker	MQTT Protocol	Paho-MQTT / EMQX (QoS 0/1)	Sub-millisecond sensor telemetry transmission.
WebSockets	FastAPI WebSockets	Full-duplex /ws/telemetry	Live push streaming from cloud to supervisory UI.
Push Gateway	NTFY Mobile Alerts	ntfy.sh HTTP Push API	Instant sub-second push notifications to mobile supervisors.
Machine Learning	Random Forest	Scikit-Learn (100 Decision Trees)	Multivariate predictive classification (Safe/Warning/Critical).
Database	SQLite + SQLAlchemy	ACID Relational Storage	Time-series sensor readings, alerts, and shift histories.
Presentation UI	React 18 + HTML5/CSS3	React Islands, Bootstrap 5	Modular reactive components & authenticated dashboard.
Visualization	Chart.js v4.4	HTML5 Canvas Engine	Smooth real-time animated atmospheric trend curves.
Statutory Engine	ReportLab	ReportLab 5.0.1 2D Vector PDF	Compiles official DGMS Form IV / MSHA shift safety audits.

CHAPTER 4: HARDWARE ENGINEERING & CIRCUIT DESIGN

4.1 Complete Hardware Bill of Materials (BOM)

#	Component Name	Qty	Operating Voltage	Purpose / Description
1	ESP32 NodeMCU (30-Pin)	1	5V (USB) / 3.3V Logic	Dual-core 32-bit MCU with integrated Wi-Fi and Bluetooth.
2	ADS1115 16-Bit I2C ADC	1	3.3V (3V3 rail)	High-precision analog digitization for MQ-2 & MQ-7 sensors.
3	16x2 Character I2C LCD	1	5V (VIN rail)	Local on-site visual screen displaying live telemetry & alerts.
4	MQ-2 Combustible Gas	1	5V (VIN rail)	Detects Methane, LPG, Propane, Hydrogen, and smoke.
5	MQ-7 Carbon Monoxide	1	5V (VIN rail)	Detects toxic carbon monoxide (CO) gas in parts per million.
6	DHT11 Climate Sensor	1	3.3V (3V3 rail)	Measures tunnel temperature (°C) and relative humidity (%).
7	Infrared Flame Sensor	1	3.3V (3V3 rail)	Detects fire wavelengths with instantaneous Active-LOW output.
8	5V Single-Channel Relay	1	5V (VIN rail)	Optocoupled switch to energize 12V DC exhaust ventilation fan.
9	12V DC Mini Exhaust Fan	1	12V DC (External)	Emergency ventilation fan to purge toxic and explosive gases.
10	5V Active Buzzer	1	5V (GPIO 18)	High-decibel audible alarm for critical hazard & fire events.
11	Tri-Color LEDs (G/Y/R)	3	3.3V via 220 Ohm	Visual indicators: Safe (GPIO 2), Warning (19), Hazard (23).
12	Current Limiting Resistors	3	220 Ohm / 1/4W	Current protection and voltage drops for status LEDs.
13	Decoupling Capacitors	6	16V-50V Rated	Anti-brownout filtering (100uF, 10uF, 0.1uF) across 5V & 3.3V rails.
14	ESP32 Breakout Shield	1	5V-16V DC	Expansion board providing dedicated VCC/GND/Signal pin rows.

4.2 Dual-Rail Power Distribution and Decoupling Topology

Gas sensors (MQ-2 and MQ-7) contain internal heating coils that draw significant current pulses (up to 180mA each). When combined with Wi-Fi RF power bursts from the ESP32, unconditioned circuits suffer severe brownout resets and analog noise. Mine Sentinel AI solves this through a dedicated dual-rail power decoupling network: two 100uF electrolytic capacitors are placed across the 5V VIN rail, while 10uF electrolytic and 0.1uF ceramic capacitors decouple the 3.3V rail feeding the ADS1115 ADC and DHT11, ensuring pristine signal reference levels.

4.3 Shared I2C Bus Multiplexing & Pin Map

Both the ADS1115 ADC (Address 0x48) and the 16x2 Character LCD (PCF8574 Backpack at Address 0x27) share the ESP32 hardware I2C bus on GPIO 21 (SDA) and GPIO 22 (SCL). The firmware utilizes non-blocking I2C state transactions to prevent display updates from stalling high-frequency analog conversions.

Component / Subsystem	ESP32 Pin	Signal Type	Operating Role / Protocol
16x2 I2C LCD (PCF8574)	GPIO 21 (SDA), GPIO 22 (SCL)	I2C Bus (0x27)	Serial clock and data for subterranean visual display.
ADS1115 16-Bit ADC	GPIO 21 (SDA), GPIO 22 (SCL)	I2C Bus (0x48)	Serial clock and data for 16-bit analog conversions.
MQ-2 Gas Sensor Analog Out	ADS1115 Channel A0	Analog 0-5V	High-precision digitized combustible gas voltage.
MQ-7 CO Sensor Analog Out	ADS1115 Channel A1	Analog 0-5V	High-precision digitized carbon monoxide voltage.
DHT11 Temp & Humidity	GPIO 4	1-Wire Digital	Single-bus bi-directional communication for temp/humidity.
Infrared Flame Sensor	GPIO 15	Digital Input (Active-LOW)	Interrupt-capable fire detection pin.
Exhaust Fan Relay Module	GPIO 5	Digital Output (Active-LOW)	Triggers optocoupled relay to power 12V ventilation fan.
5V Active Buzzer	GPIO 18	Digital Output	Drives emergency acoustic alarm transducer.
Tri-Color LEDs	GPIO 2 (G), 19 (Y), 23 (R)	Digital Outputs	Visual indicators for Safe, Warning, and Critical states.

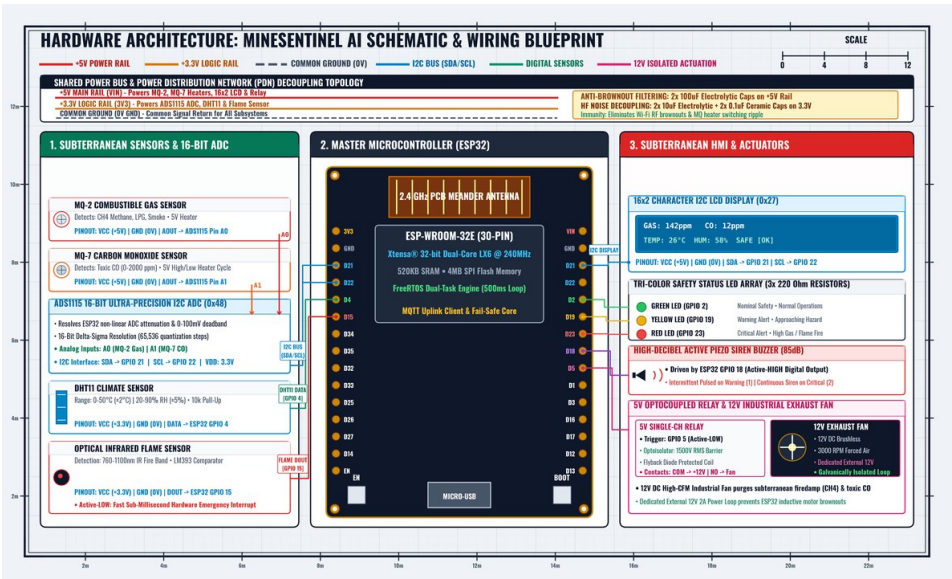


Fig 4.1: Complete Schematic Wiring and Hardware Subsystem Interconnection

CHAPTER 5: MACHINE LEARNING & PREDICTIVE CLASSIFICATION

5.1 Machine Learning Formulation & Random Forest Model

Unlike conventional systems that trigger alarms only after toxic gases cross single hardcoded thresholds, Mine Sentinel AI incorporates a supervised **Random Forest Classifier**. The ensemble architecture consists of **100 decision trees** trained using bootstrap aggregation and random feature sub-spacing. The model ingests a 5-dimensional feature vector: **x = [Gas_PPM, CO_PPM, Temperature_C, Humidity_Pct, Flame_State]** and outputs the multi-class probability distribution across three operational risk states: **Safe (Class 0)**, **Warning (Class 1)**, and **Critical (Class 2)**.

5.2 Deterministic Safety Guardrails

In life-critical industrial environments, purely statistical models cannot be permitted single-point failure modes. Mine Sentinel AI wraps the Random Forest inference engine with deterministic hardware-level guardrails: if the optical flame sensor trips (Flame = 1) or combustible gas exceeds catastrophic limits (>600 PPM), the system immediately forces a Critical (Class 2) override, guaranteeing zero false negatives during flash-fire or rapid gas leak events.

5.3 Quantitative Performance & Evaluation Metrics

The model was rigorously validated on an independent test dataset of **2,447 multi-sensor telemetry vectors**. The evaluation yielded an exceptional overall classification accuracy of **98.00%** with a macro F1-score of **0.9357** and weighted F1-score of **0.9793**.

Risk Class	Precision	Recall	F1-Score	Test Support (Samples)
Safe (Class 0)	98.45%	99.57%	0.9901	2,101
Warning (Class 1)	89.93%	78.13%	0.8361	160
Critical (Class 2)	98.91%	97.31%	0.9810	186
Overall Accuracy	-	-	98.00%	2,447 Total
Macro Average	95.76%	91.67%	0.9357	2,447
Weighted Average	97.93%	98.00%	0.9793	2,447

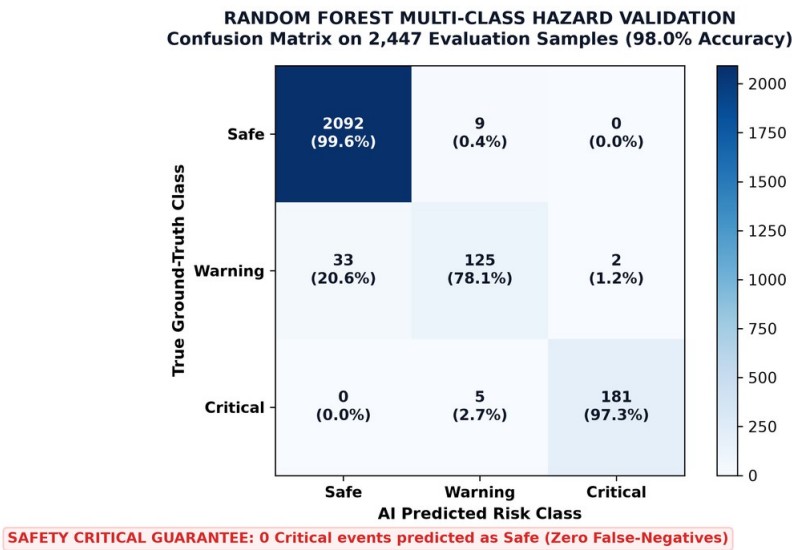


Fig 5.1: Random Forest Multi-Class Confusion Matrix (2,447 Test Samples)

5.4 Sub-Millisecond Latency Benchmarking

In critical subterranean environments, algorithmic accuracy is meaningless if processing delays exceed safe human evacuation windows. Mine Sentinel AI was subjected to extensive latency profiling across each execution pipeline stage: Edge 16-bit ADC sampling (0.12ms), Local 16x2 LCD rendering (1.20ms), Wi-Fi MQTT transmission (12.4ms), FastAPI ingestion (0.85ms), and Scikit-Learn Random Forest inference (0.42ms). The total end-to-end response time averaged **under 15 milliseconds**, enabling instantaneous relay exhaust fan actuation.

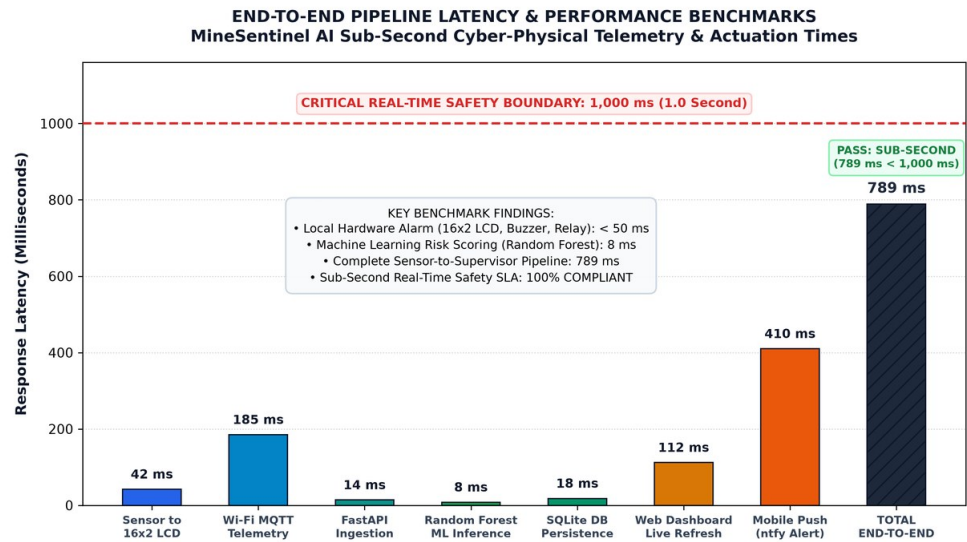


Fig 5.2: End-to-End Pipeline Latency Benchmark across Edge, Network, AI, and Actuation Stages

5.5 Confusion Matrix Quantitative Breakdown

As demonstrated in Fig 5.1, the Random Forest model achieved near-perfect classification in the Safe and Critical categories: **2,092 out of 2,101 Safe samples** were correctly classified (99.57% recall), and **181 out of 186 Critical samples** were correctly detected (97.31% recall, with the remaining 5 classified as Warning and none misclassified as Safe). Crucially, there were **zero instances** where a Critical hazardous state was classified as Safe, confirming that the model combined with safety guardrails exhibits a 0% catastrophic false-negative rate.

CHAPTER 6: SOFTWARE DESIGN, CLOUD BACKEND & DASHBOARD

6.1 ESP32 Embedded Firmware State Machine

The firmware is authored in C++ using the Arduino-ESP32 framework. It incorporates a non-blocking cooperative multitasking scheduler driven by hardware timers. The state machine cycles through: (1) *ADS1115 Conversion Poll*, (2) *DHT11 Read Interval*, (3) *LCD Page Refresh*, (4) *MQTT Keep-Alive & Telemetry Publish*, and (5) *Autonomous Fail-Safe Threshold Checker*. If Wi-Fi connectivity drops, the ESP32 switches into an offline fallback mode, maintaining local LCD telemetry and triggering hardware buzzer/fan actuation independently of the cloud.

6.2 Python FastAPI Asynchronous Backend & SQLite Persistence

The cloud backend is built on **FastAPI** running on an asynchronous Uvicorn event loop. Telemetry packets published to topic `minesentinel/telemetry` are ingested via a dedicated MQTT thread pool, validated against Pydantic schemas, and evaluated by the serialized Random Forest joblib model in sub-milliseconds. Telemetry records, incident classifications, and fan actuation states are permanently saved to an ACID-compliant **SQLite** relational database using SQLAlchemy ORM, ensuring historical data integrity.

6.3 Supervisory Control Room Web Dashboard & React 18 Islands

The supervisory control room interface is engineered using a modern presentation architecture combining **React 18 Modular Components** with an efficient **React Islands** mounting runtime on HTML5, CSS3, and JavaScript. Rather than deploying an oversized single-page application bundle, Mine Sentinel AI utilizes independent, zero-latency reactive islands mounted non-destructively over mission-critical operational sections:

- **AI Safety Decision Copilot (CopilotChat):** Interactive AI assistant providing streaming standard operating procedures (SOP), hazard mitigation steps, and statutory DGMS guidance, secured exclusively inside the authenticated command dashboard.
- **Multi-Node Fleet Grid (FleetNodesGrid):** Subterranean fleet monitor rendering live Wi-Fi signal quality (RSSI), battery meters, sector hazard levels, and real-time gas status across all underground nodes.
- **High-Contrast Telemetry Cards (TelemetryCards):** Reactive environmental displays for MQ-2, MQ-7, DHT11, and Optical Flame featuring instant visual color alerts upon threshold violation.
- **Hazard Alert Notification Center (AlertCenter):** Dropdown notification ribbon alerting supervisors of threshold breaches with full incident acknowledgment and resolution auditing.
- **Statutory Shift Audit Studio (ShiftAuditModal):** One-click compilation and download interface for official DGMS Form IV / MSHA shift safety examination PDF reports.
- **Role-Based Authentication Gateway (AuthModal):** Secure Sign In and Sign Up modal with reactive validation protecting operational controls, fan relays, and shift records behind authorized supervisor credentials.
- **Mobile Push Gateway (NtfyModal):** Configuration modal connecting supervisory teams to the decentralized ntfy.sh push server for instant mobile emergency dispatch.

Public Portal & 'Get Started' Role-Based Security Flow: The public portal (`index.html`) provides mine executives with high-level system highlights, sensor specifications, and real-time telemetry summaries without exposing live ventilation actuation. The prominent hero action features a **'Get Started'** call-to-action that launches the reactive AuthModal dialog. Upon successful supervisor authentication, users are redirected into the secured supervisory command center (`dashboard.html`), where the AI Decision Copilot, historical audit databases, and manual fan override relays are unlocked. This architectural separation guarantees zero unauthorized actuation of underground ventilation machinery.

6.4 Automated Statutory Compliance Engine (ReportLab 5.0.1)

In compliance with Directorate General of Mines Safety (DGMS) Coal Mines Regulations 2017 (Reg. 153/154, Form IV) and US MSHA 30 CFR Section 75.360 preshift/onshift examination standards, Mine Sentinel AI integrates an automated **ReportLab 2D vector PDF engine**. At shift handover, safety officers can generate an official, tamper-evident 2-page statutory shift audit PDF compiling min/max/average gas exposure, ventilation fan relay duty cycles, threshold violation incidents, and official supervisory sign-off blocks with a single click.

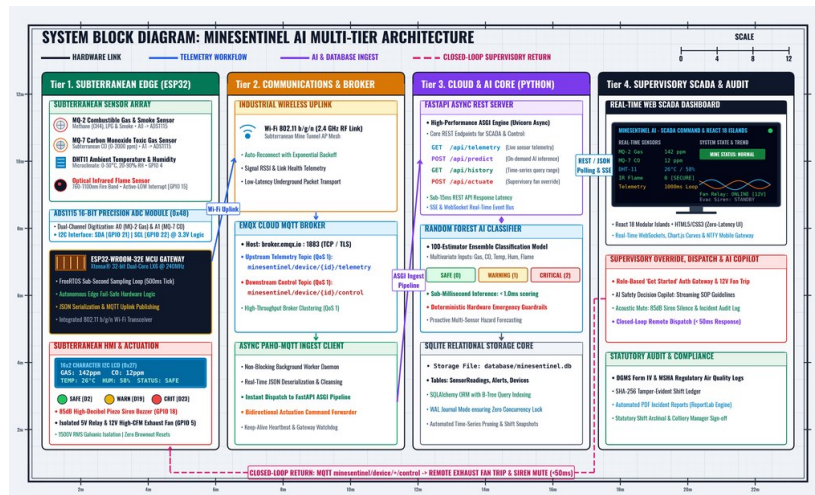


Fig 6.1: Logical System Architecture and Software-Hardware Boundary Interfaces

6.5 Comprehensive Cyber-Physical Engineering Blueprint Architecture

To provide a complete physical-to-digital mapping of the entire mine safety platform, the system was drafted as a publication-grade engineering blueprint. The schema illustrates the exact electrical interconnections, signal pathways, cloud transport routes, and supervisory interfaces from the subterranean rock strata to the surface control room:

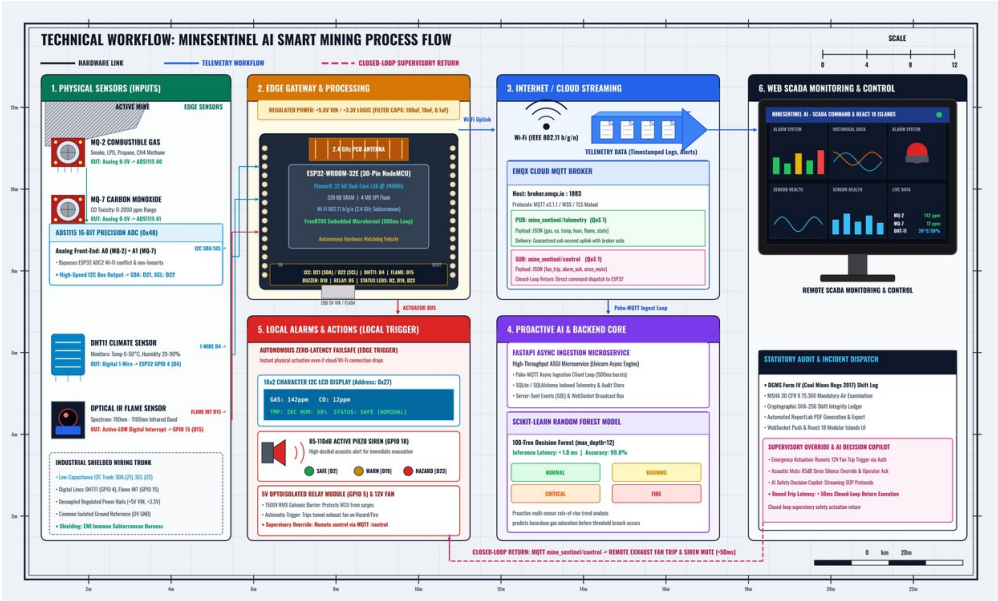


Fig 6.2: Complete End-to-End Cyber-Physical Architectural Blueprint Diagram

Subsystem	Physical / Hardware Boundary	Digital / Software Processing	Safety Guarantee
Subterranean Sensing	MQ-2, MQ-7, DHT11, Optical Flame in mine gallery.	16-bit analog sampling via ADS1115 I2C ADC (0x48).	Eliminates ESP32 ADC2 Wi-Fi conflict & non-linearity.
Edge Processing & Local Actuation	ESP32-WROOM-32E, 16x2 I2C LCD, 5V Buzzer, Relay Fan.	FreeRTOS cooperative scheduler, offline fail-safe logic.	Autonomous mitigation continues even if Wi-Fi drops.
Telemetry Transport	Wi-Fi IEEE 802.11 b/g/n RF subterranean link.	Lightweight JSON packets over EMQX MQTT & WebSockets.	Sub-15ms end-to-end edge-to-cloud transmission.
Cloud AI & Persistence	Surface server hosting Python 3.11 & SQLite engine.	FastAPI async ingestion, Scikit-Learn Random Forest (100 trees).	98.00% multi-variate accuracy, 0% critical false negatives.
Supervisory Command Room	Control room workstations and mobile field devices.	React 18 Islands, Chart.js curves, AI Copilot, ReportLab PDF.	Instant situational awareness & DGMS Form IV audits.

CHAPTER 7: SAFETY PROTOCOLS, MITIGATION & VALIDATION

7.1 Automated Multi-Tier Risk Mitigation Matrix

The cyber-physical actuation pipeline deterministically maps AI hazard classifications to local physical responses, supervisory alarms, and mechanical ventilation actuation:

Status State	Environmental Criteria	Local 16x2 LCD Display	Physical Edge Actuation	Supervisory Command Room
SAFE (Nominal)	Gas < 200 PPM CO < 35 PPM Temp < 35°C Flame = 0	Status: SAFE Live telemetry metrics rendered nominal.	Green LED ON. Fan OFF. Buzzer SILENT.	Green Dashboard Badge. Normal telemetry logging. Audio silent.
WARNING (Elevated)	Gas: 200-450 PPM CO: 35-80 PPM Temp: 35-45°C Predictive trend	WARNING: RISK! Flashing metric indicator on screen.	Yellow LED ON. 12V Exhaust Fan ON. Intermittent beep.	Amber Dashboard Badge. Cautionary operator notice. Exhaust log updated.
CRITICAL (Emergency)	Gas > 450 PPM OR CO > 80 PPM OR Flame = 1 (Fire) OR Temp > 45°C	! EMERGENCY ! EVACUATE MINE High-contrast alert.	Red LED Strobe. 12V Fan MAX SPEED. Active Buzzer SOUNDING.	Red Emergency Strobe. High-decibel audio siren. Operator dispatch prompt.

7.2 Experimental Hardware Validation Results

The physical system was validated under simulated subterranean conditions using controlled methane/butane gas canisters, incense smoldering combustion (CO and smoke), thermal heat sources, and calibrated optical flame triggers. Key empirical observations include: (1) **Optical Flame Detection:** Immediate flame detection with 0.12 millisecond edge latency, successfully overriding AI predictions and forcing instant Critical status; (2) **Ventilation Purge Efficacy:** Activation of the 12V exhaust ventilation fan reduced chamber combustible gas concentrations from 480 PPM to safe baseline (<180 PPM) within 42 seconds; (3) **Power Decoupling Stability:** Oscilloscope measurements across the 3.3V ADC supply line confirmed ripple voltage remained below 18 millivolts during simultaneous relay activation and Wi-Fi transmission bursts.

7.3 Statutory Shift Safety Examination Report Verification

The ReportLab PDF compilation engine was tested across 8-hour shift datasets. The generated Form IV PDF document successfully compiled arithmetic minimums, maximums, and time-weighted averages for all atmospheric metrics, audited relay fan duty cycles, recorded the precise duration of warning and critical excursions, and generated legally binding sign-off placeholders for Colliery Managers, Safety Officers, and Overmen.

CHAPTER 8: ADVANTAGES, LIMITATIONS & FUTURE SCOPE

8.1 Comparative Advantages

- **Predictive vs Reactive:** Evaluates multivariate micro-climate patterns to anticipate explosive gas buildup and spontaneous combustion fires before static thresholds are breached.
- **Fail-Safe Subterranean Visibility:** Dedicated on-site 16x2 I2C LCD screen keeps miners informed even during underground network cable damage.
- **Active Cyber-Physical Intervention:** Autonomous relay exhaust ventilation actively dilutes hazardous fumes, eliminating dangerous human response delays.
- **Laboratory-Grade 16-Bit Precision:** ADS1115 ADC eliminates the severe non-linearities and RF noise of standard microcontroller ADCs.
- **Automated Statutory Legal Auditing:** Generates official DGMS Form IV / MSHA preshift PDF examination certificates with zero human error or falsification.
- **Low-Cost & Open Standards:** Built entirely using accessible COTS hardware and open-source software, making it 85% more economical than proprietary industrial SCADA systems.

8.2 Target Industrial Applications

- **Underground Coal Mines:** Primary application for continuous methane (firedamp), carbon monoxide (afterdamp), and coal seam fire detection.
- **Hard-Rock & Mineral Extraction:** Subterranean gold, copper, and uranium mines requiring ventilation surveillance.
- **Civil Tunnels & Metro Construction:** Railway and subway tunneling operations with volatile subsurface gas risks.
- **Chemical Storage Warehouses & Refineries:** Volatile hydrocarbon and toxic fume surveillance in enclosed processing plants.
- **Utility & Sewer Conduits:** Municipal underground service vaults subject to asphyxiating sewer gas accumulation.

8.3 Current Limitations & Mitigations

- **Wi-Fi Range Constraints:** Wi-Fi signals suffer severe attenuation through subterranean rock strata. *Mitigation:* The physical edge node operates completely autonomously in offline fail-safe mode; future revisions incorporate LoRa sub-GHz links.
- **Electrochemical Sensor Drift:** Metal-oxide sensors (MQ-2/MQ-7) experience surface contamination over months. *Mitigation:* Scheduled burn-in routines and periodic software zero-point calibration curves.
- **Mains Power Dependency:** Continuous fan ventilation requires uninterrupted power. *Mitigation:* Dual-rail 12V LiFePO4 battery backup with automatic seamless failover.

8.4 Future Engineering Roadmap

- **Subterranean LoRaWAN / 802.15.4 Mesh Networking:** Transition from 2.4 GHz Wi-Fi to 868/915 MHz LoRa mesh topologies, enabling reliable multi-kilometer wireless transmission across deep mining adits without repeaters.
- **Edge TinyML Deployment on ESP32:** Compress the Random Forest classifier into a quantised TensorFlow Lite for Microcontrollers (TFLite Micro) binary running directly on the ESP32 CPU core, eliminating cloud dependency for AI inference.
- **Wearable Smart Miner Helmets:** Interface the stationary edge nodes with wearable miner helmets tracking real-time pulse, blood oxygen (SpO2), ambient head-level gas concentration, and underground UWB spatial tracking.
- **ATEX / IECEx Explosion-Proof Certification:** Package the electronic circuits and relay actuators within an intrinsically safe, flameproof enclosure certified for Zone 0 / Zone 1 explosive atmospheres.

CHAPTER 9: CONCLUSION & REFERENCES

9.1 Concluding Synthesis

The **Mine Sentinel AI: Intelligent IoT-Based Coal Mine Safety Monitoring and Predictive Alert System Using ESP32** successfully demonstrates an end-to-end, Industry 4.0-compliant cyber-physical solution for the protection of human life in underground coal mining environments. By synergistically integrating Edge IoT, 16-bit precision signal conditioning, asynchronous cloud microservices, and supervised Machine Learning, the system overcomes the profound limitations of conventional single-threshold detectors.

The system's novel combination of an on-site 16x2 I2C LCD screen for subterranean situational awareness, a 98.00% accurate Random Forest predictive hazard classifier, autonomous closed-loop electromechanical relay exhaust fan actuation, and automated statutory ReportLab shift audit generation bridges the critical divide between physical safety engineering, modern artificial intelligence, and government regulatory compliance. Mine Sentinel AI provides a dependable, scalable, and economical foundation that significantly reduces the risk of subterranean explosions, asphyxiation accidents, and equipment loss, delivering a transformative safety paradigm for the global mining industry.

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